# Functional Annotation Report: prpB (Q88KF6) — 2-Methylisocitrate Lyase from *Pseudomonas putida* KT2440

## Summary

**prpB** (ordered locus **PP_2334**; synonym *mmgF*; UniProt **Q88KF6**) of *Pseudomonas putida* KT2440 encodes **2-methylisocitrate lyase** (MICL, also abbreviated 2-MIC; **EC 4.1.3.30**). This is a **$Mg^{2+}$-dependent carbon–carbon lyase** that catalyzes the **terminal, committed step of the 2-methylcitrate cycle**: the cleavage of **(2R,3S)-2-methylisocitrate into succinate and pyruvate**. The 2-methylcitrate cycle is the principal aerobic route by which bacteria oxidize and detoxify **propionate/propionyl-CoA**, the three-carbon acyl unit generated during catabolism of odd-chain-length fatty acids, branched-chain amino acids, and related substrates. By splitting 2-methylisocitrate, PrpB channels odd-chain carbon into the TCA cycle (via succinate) and central metabolism (via pyruvate).

PrpB is a **cytoplasmic** enzyme, consistent with its role in a soluble intermediary-metabolism pathway alongside the other Prp enzymes (methylcitrate synthase PrpC, methylcitrate dehydratase PrpD, and the aconitase-like AcnD/PrpF activity). Structurally, PrpB belongs to the **isocitrate lyase / phosphoenolpyruvate (PEP) mutase superfamily**, adopting a TIM-barrel $(\beta/\alpha)_8$ fold and typically assembling into a **homotetramer** of ~32 kDa subunits. Catalysis proceeds through a mobile **"active-site loop"** that closes over the bound substrate to exclude solvent, and the reaction runs via an **α-carboxy-carbanion intermediate** with inversion of configuration at C3 of 2-methylisocitrate. Two residues — **cysteine 123 and aspartate 58** (Salmonella numbering) — are essential for catalysis, and $Mg^{2+}$ is strictly required to coordinate the pyruvate product/enolpyruvate intermediate. Substrate specificity is strict: the closely related molecule isocitrate binds but acts as a **dead-end inhibitor** rather than a substrate.

The identity of Q88KF6 as prpB/2-methylisocitrate lyase is **well supported and unambiguous**. The gene symbol, protein family, and catalytic domains from UniProt align precisely with an extensive body of biochemical and structural literature on PrpB in *Salmonella enterica*, *Escherichia coli*, and *Pseudomonas* species. While the deepest mechanistic characterization

comes from enterobacterial orthologs, the *Pseudomonas*-specific literature directly confirms both the identity and the physiological importance of PrpB for propionate oxidation, including downstream consequences for polyhydroxyalkanoate (PHA) biopolymer composition.

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## Key Findings

### Finding 1 — PrpB is a $Mg^{2+}$-dependent 2-methylisocitrate lyase catalyzing the terminal step of the 2-methylcitrate cycle

PrpB (EC 4.1.3.30) catalyzes the **retro-aldol cleavage of (2R,3S)-2-methylisocitrate into succinate + pyruvate**, which is the **last step of the 2-methylcitrate cycle** for propionate oxidation. This reaction is directly analogous to the isocitrate-lyase reaction of the glyoxylate shunt (which cleaves isocitrate into succinate + glyoxylate), but with strict selectivity for the methyl-branched substrate.

The most precise kinetic characterization comes from the *Salmonella enterica* ortholog, whose properties are expected to closely mirror those of the *P. putida* enzyme given their shared family and reaction. In *S. enterica*, PrpB acts on 2-methylisocitrate with a **$K_m$ of 19 µM and a kcat of 105 $s^{-1}$**, indicating a high-affinity, catalytically efficient enzyme PMID: 12897003. The enzyme has a **strict $Mg^{2+}$ requirement** ($Mn^{2+}$ substitutes poorly, giving only ~28% of activity), a pH optimum near 7.5, a temperature optimum around 50 °C, and assembles as a **homotetramer with ~32 kDa subunits**. Crystal structures of the *E. coli* and *S. typhimurium* enzymes confirm the reaction chemistry and reveal how the pyruvate product is coordinated by the essential divalent metal ion PMID: 12706720; PMID: 14575713.

> "The K(m) of PrpB for 2-MIC was measured at 19 micro M, with a k(cat) of 105 s(-1)." — PMID: 12897003
>
> "the cleavage of 2-methylisocitrate to succinate and pyruvate, is catalysed by 2-methylisocitrate lyase (PrpB)" — PMID: 14575713

The reaction PrpB catalyzes can be written as:

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(2R,3S)-2-methylisocitrate  --(PrpB, Mg2+)-->  succinate + pyruvate
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This positions PrpB as the **carbon-releasing exit valve** of the methylcitrate cycle: the two products feed directly into central carbon metabolism, with succinate entering the TCA cycle and pyruvate available for oxidation, gluconeogenesis, or biosynthesis.

## Finding 2 — PrpB belongs to the ICL/PEP-mutase superfamily and uses active-site-loop closure with an α-carboxy-carbanion mechanism; C123 and D58 are catalytically essential

PrpB adopts the **isocitrate lyase / PEP mutase (ICL/PEPM) fold**, a variant of the TIM-barrel architecture shared across a superfamily that includes isocitrate lyase, phosphoenolpyruvate mutase, and carboxyphosphonoenolpyruvate mutase. A defining mechanistic feature of this superfamily — directly observed for PrpB — is the movement of a **mobile "active-site loop"** from an open to a closed conformation upon substrate binding, sequestering the reactants from bulk solvent during catalysis PMID: 12706720; PMID: 15723538.

The detailed catalytic mechanism was resolved through structures of an active-site **C123S variant** captured in complex with product and with the inhibitor isocitrate. These structures support a mechanism proceeding through an **α-carboxy-carbanion intermediate / transition state**, consistent with prior stereochemical experiments demonstrating **inversion of configuration at C3** of 2-methylisocitrate. Crucially, isocitrate — which lacks the C2 methyl group but is otherwise similar — binds in the active site as a **dead-end inhibitor**, explaining the enzyme's strict substrate and stereochemical selectivity PMID: 15723538. Site-directed mutagenesis independently identified **Cys123 and Asp58** as residues essential for catalysis PMID: 12897003.

> "We propose a catalytic mechanism involving an alpha-carboxy-carbanion intermediate/transition state, which is consistent with previous stereochemical experiments showing inversion of configuration at the C(3) of 2-methylisocitrate." — PMID: 15723538
>
> "A common feature of members of this enzyme family is the movement of a so-called 'active site loop' from an open into a closed conformation upon substrate binding thus shielding the reactants from the surrounding solvent." — PMID: 12706720

Together these studies establish PrpB as a mechanistically well-understood lyase whose fold, metal dependence, and residue-level chemistry are conserved features of the ICL/PEPM superfamily. The strict discrimination between 2-methylisocitrate (substrate) and isocitrate (inhibitor) is the structural basis for keeping the methylcitrate cycle biochemically distinct from the glyoxylate shunt, despite their chemical parallels.

### Finding 3 — In *Pseudomonas*, PrpB is required for propionate oxidation via the only known propionate pathway, with consequences for PHA composition

The physiological importance of PrpB in *Pseudomonas* is directly documented. In *Pseudomonas* sp. LFM693, a **prpB (2-methylisocitrate lyase) disruption mutant** blocks the 2-methylcitrate cycle — explicitly described as **the only known and characterized pathway for propionate oxidation in this organism**. With the cycle interrupted, propionyl-CoA is no longer fully oxidized and is instead redirected into **odd-chain-length polyhydroxyalkanoate (PHA) monomers** and accumulating 2-methylisocitrate, providing a whole-cell readout of PrpB's metabolic role PMID: 37995793; PMID: 39492536.

> "Pseudomonas sp. LFM693 is a 2-methylisocitrate lyase (prpB) disrupted mutant. This enzyme catalyzes a step in the 2-methylcitrate cycle, the only known and described pathway for propionate oxidation in this organism." — PMID: 37995793

In the closely related *P. aeruginosa*, the 2-methylcitrate and glyoxylate cycles are further shown to be **linked through enzymatic redundancy**: the glyoxylate-cycle isocitrate lyase **AceA exhibits secondary 2-methylisocitrate lyase activity**, demonstrated through structural analysis, kinetics, metabolomics, and PrpB mutation. This means that in some *Pseudomonas* backgrounds a portion of MICL activity can be supplied by the bifunctional ICL enzyme, echoing the situation in *Mycobacterium tuberculosis*, where ICL isoforms substitute for a missing dedicated MICL PMID: 40015638; PMID: 18048912.

> "isocitrate lyase (AceA) exhibiting secondary 2-methylisocitrate lyase activity" — PMID: 40015638

These functional-genetics results confirm the biochemical annotation at the physiological level: PrpB is the dedicated MICL that enables *Pseudomonas* to grow on and detoxify propionate, and its loss has measurable, characteristic metabolic consequences.

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# Mechanistic Model / Interpretation

## The 2-methylcitrate cycle and PrpB's place in it

Propionate (or propionyl-CoA generated internally) is oxidized aerobically through the **2-methylcitrate cycle**, a pathway analogous to the citric-acid and glyoxylate cycles but adapted for the three-carbon propionyl unit. PrpB catalyzes the final, carbon-releasing step:

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propionate
    │  (PrpE / acyl-CoA synthetase)
    ▼
propionyl-CoA  + oxaloacetate
    │  PrpC  (2-methylcitrate synthase)
    ▼
(2S,3S)-2-methylcitrate
    │  PrpD / AcnD  (2-methylcitrate dehydratase)
    ▼
2-methyl-cis-aconitate
    │  PrpF / AcnB  (aconitase-type hydratase/isomerase)
    ▼
(2R,3S)-2-methylisocitrate
    │  ★ PrpB  (2-METHYLISOCITRATE LYASE)  ── requires Mg2+
    ▼
SUCCINATE  +  PYRUVATE
    │                 │
    ▼                 ▼
TCA cycle       central metabolism
(regenerates    (oxidation, gluconeogenesis,
 oxaloacetate)    biosynthesis)
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The succinate product re-enters the TCA cycle and ultimately regenerates the oxaloacetate consumed by PrpC, making the pathway catalytically self-sustaining; the net effect is the complete oxidation of propionate while the pyruvate output supplies carbon to the rest of the cell. PrpB is therefore both the **exit point** of the cycle and a **detoxification valve**: without it, propionyl-CoA and pathway intermediates accumulate and can be toxic, as seen in mycobacterial models where blocking MICL activity is deleterious PMID: 18048912.

## Structural/mechanistic model

| Property | Value / description | Evidence |
|---|---|---|
| EC number | 4.1.3.30 | UniProt / PMID: 14575713 |
| Reaction | (2R,3S)-2-methylisocitrate → succinate + pyruvate | PMID: 14575713 |
| Metal cofactor | $Mg^{2+}$ required ($Mn^{2+}$ ~28% activity) | PMID: 12897003 |
| $K_m$ (2-MIC) | 19 µM (*S. enterica*) | PMID: 12897003 |
| kcat | 105 $s^{-1}$ (*S. enterica*) | PMID: 12897003 |
| pH / T optimum | ~7.5 / ~50 °C | PMID: 12897003 |
| Quaternary structure | Homotetramer, ~32 kDa/subunit | PMID: 12897003 |
| Fold / superfamily | ICL/PEPM TIM-barrel; isocitrate lyase/PEP mutase superfamily | PMID: 12706720 |
| Catalytic mechanism | α-carboxy-carbanion intermediate; inversion at C3 | PMID: 15723538 |
| Essential residues | Cys123, Asp58 (Salmonella numbering) | PMID: 12897003 |
| Substrate specificity | Strict for 2-methylisocitrate; isocitrate is a dead-end inhibitor | PMID: 15723538 |
| Subcellular localization | Cytoplasm | Inferred from soluble intermediary-metabolism pathway; consistent with family |

Catalysis is a coordinated cycle: the substrate binds in the TIM-barrel active site, the **mobile loop closes** to exclude water, $Mg^{2+}$ stabilizes the developing negative charge, the **C–C bond is cleaved** through an α-carboxy-carbanion (enolpyruvate-like) transition state with inversion at C3, and the loop reopens to release succinate and pyruvate. The methyl branch of the substrate is essential for productive binding — its absence (i.e., isocitrate) converts the ligand into a non-cleavable inhibitor, which is how the cell keeps the methylcitrate and glyoxylate cycles chemically insulated even though both use ICL-family folds.

## Localization

PrpB carries out its function in the **cytoplasm**. The 2-methylcitrate cycle is a soluble intermediary-metabolism pathway; its enzymes (PrpC, PrpD, AcnD/PrpF, PrpB) act on small polar organic acids and CoA thioesters within the cytosol, and PrpB has no signal peptide or membrane-anchoring features characteristic of secreted or membrane proteins. Its products (succinate, pyruvate) are cytosolic metabolites feeding the central carbon network.

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## Evidence Base

| PMID | Title (abbrev.) | Organism | Contribution |
|---|---|---|---|
| 12897003 | Residues C123 and D58 of PrpB essential for catalysis | *S. enterica* | Kinetics ($K_m$, kcat), $Mg^{2+}$ dependence, quaternary structure, essential residues |
| 12706720 | Crystal structure of *E. coli* PrpB | *E. coli* | Fold, ICL/PEPM superfamily membership, active-site-loop closure |
| 14575713 | Crystal structure of *S. typhimurium* PrpB + pyruvate/$Mg^{2+}$ | *S. typhimurium* | Reaction definition; $Mg^{2+}$-pyruvate coordination |
| 15723538 | PrpB with product and isocitrate inhibitor | *E. coli/ Salmonella* | α-carboxy-carbanion mechanism; C3 inversion; substrate specificity |
| 37995793 | Fed-batch cultivation; PHA composition | *Pseudomonas* sp. LFM693 | prpB mutant; MICL as only propionate-oxidation route |
| 39492536 | Propionate oxidation & odd-chain PHA | *Pseudomonas* sp. LFM046 | Physiological role; propionyl-CoA rerouting on MICL loss |
| 40015638 | Methylcitrate/glyoxylate cycles linked by redundancy | *P. aeruginosa* | AceA (ICL) has secondary MICL activity; PrpB mutation |
| 18048912 | Methylcitrate cycle in propionate detoxification | *M. smegmatis* | prpB clustering with prpC/prpD; detoxification role; ICL redundancy |
| 30745367 | Nitrogen regulator GlnR controls transcription | (regulatory) | Context for prp operon transcriptional control |

**How the evidence fits together.** The four enterobacterial structural/biochemical papers (PMID: 12897003, 12706720, 14575713, 15723538) provide the gold-standard molecular characterization of PrpB — the reaction, kinetics, metal dependence, fold, mechanism, essential residues, and substrate specificity. Because *P. putida* PrpB shares the same gene symbol, EC number, family, and diagnostic domains (ICL/PEPM, PrpB IPR012695), these mechanistic conclusions transfer with high confidence. The *Pseudomonas*-specific papers

(PMID: 37995793, 39492536, 40015638) confirm the physiological role in this genus and reveal the AceA/PrpB redundancy that connects the methylcitrate and glyoxylate cycles. The mycobacterial paper (PMID: 18048912) reinforces the detoxification theme and the recurring evolutionary theme of ICL/MICL functional overlap. No cited evidence contradicts the annotation; all sources converge on the same identity and function.

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## Limitations and Knowledge Gaps

1. **No *P. putida* KT2440–specific enzymology.** The most rigorous kinetic and structural data ($K_m$, kcat, $Mg^{2+}$ dependence, C123/D58, mechanism) come from *S. enterica* and *E. coli* orthologs, not from Q88KF6 itself. Transfer of these parameters relies on strong sequence/family homology rather than direct measurement on the *P. putida* protein. Absolute kinetic constants may differ modestly.
2. **Localization is inferred, not directly demonstrated.** Cytoplasmic localization is deduced from pathway biochemistry and the absence of targeting signals; no experimental localization study for the *P. putida* enzyme is cited.
3. **Residue numbering is from Salmonella.** The essential Cys123/Asp58 map onto conserved positions, but exact residue numbers in the *P. putida* sequence require alignment to confirm.
4. **Redundancy in *P. putida* is not directly established.** The AceA secondary-MICL-activity result is from *P. aeruginosa*. Whether *P. putida* KT2440's isocitrate lyase similarly backs up PrpB has not been directly demonstrated, so the strict "only known pathway" statement (from *Pseudomonas* sp. LFM693) may be organism/strain-dependent.
5. **Regulation is only lightly covered.** The transcriptional/regulatory control of the *prp* operon in *P. putida* (e.g., PrpR activation, 2-methylcitrate as inducer) was not deeply investigated here; the GlnR paper (PMID: 30745367) provides only partial context.

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## Proposed Follow-up Experiments / Actions

1. **Direct biochemical validation of Q88KF6.** Recombinantly express and purify *P. putida* KT2440 PrpB and measure $K_m$, kcat, and metal dependence on authentic (2R,3S)-2-methylisocitrate to confirm parameters transferred from *Salmonella*.
2. **Confirm essential residues by alignment + mutagenesis.** Align Q88KF6 to *S. enterica* PrpB, identify the residues corresponding to Cys123/Asp58, and verify catalytic essentiality by site-directed mutagenesis.
3. **Test genetic redundancy in KT2440.** Construct ΔprpB, ΔaceA (icl), and double mutants and assay growth on propionate (and propionate + glucose) to determine whether *P. putida* isocitrate lyase provides backup MICL activity as in *P. aeruginosa*.
4. **Metabolite and localization studies.** Use metabolomics to confirm 2-methylisocitrate accumulation in a KT2440 ΔprpB mutant, and perform fractionation or fluorescent-tagging to confirm cytoplasmic localization.
5. **Structural confirmation.** Solve or model (AlphaFold) the *P. putida* PrpB structure, verify the ICL/PEPM TIM-barrel fold and the mobile active-site loop, and dock 2-methylisocitrate to confirm the α-carboxy-carbanion mechanistic geometry.
6. **Regulatory mapping.** Characterize the *prpBCDF* operon promoter and its regulator(s) in KT2440 to define how propionate exposure induces PrpB expression.

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*Report compiled from an autonomous, literature-grounded investigation. All quantitative claims are drawn from the cited primary literature; where data derive from orthologous enzymes (chiefly* S. enterica *and* E. coli*), this is stated explicitly.*

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